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C

rate of heat dissipation $P = I_{\text{rms}}^2 R = \frac{I_o^2}{2} R$

for half-rectified wave, $I_{\text{rms}} = \frac{I_o}{2}$

so rate of heat dissipation $P_{\text{new}} = I_{\text{rms}}^2 R = \frac{I_o^2}{4} R = \frac{P}{2}$

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